

Filter banks localization

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1 EnVar formulation

1.1 Standard EnVar formulation

For an ensemble N backgrounds $\{\mathbf{x}^1, \dots, \mathbf{x}^N\}$, with $\mathbf{x}^k \in \mathbb{R}^n$, the standard EnVar formulation is given by:

$$\mathbf{B} = \mathbf{X}\mathbf{X}^T \circ \mathbf{L} \quad (1)$$

where:

- $\mathbf{B} \in \mathbb{R}^{n \times n}$ is the resulting background error covariance.
- $\mathbf{X} \in \mathbb{R}^{n \times N}$ contains the normalized ensemble perturbations:

$$X_{ij} = \frac{1}{\sqrt{N-1}} \left(x_i^j - \langle x_i \rangle \right) \quad (2)$$

with $\langle x_i \rangle$ denoting the ensemble mean:

$$\langle x_i \rangle = \frac{1}{N} \sum_{j=1}^N x_i^j \quad (3)$$

- $\mathbf{L} \in \mathbb{R}^{n \times n}$ is the localization matrix.
- $\circ$ denotes a Schur product between matrices (element-by-element).

The square-root expression of (1), transforming the control vector $\mathbf{v} \in \mathbb{R}^m$ into a control increment $\delta \mathbf{x} \in \mathbb{R}^n$, is given by:

$$\delta \mathbf{x} = \sum_{j=1}^N \mathbf{X}_{:,j} \circ \left(\mathbf{U} \mathbf{v}_j \right) \quad (4)$$

where:

- $\mathbf{X}_{:j} \in \mathbb{R}^n$ is the j^{th} column of $\mathbf{X}$
- $\mathbf{U} \in \mathbb{R}^{n \times m}$ is a square-root of the localization $\mathbf{L}$ such that $\mathbf{U}\mathbf{U}^T = \mathbf{L}$.
- $\mathbf{v}_j \in \mathbb{R}^{m/N}$ is the j^{th} subvector of $\mathbf{v}$.

1.2 Generalized EnVar formulation

The standard EnVar formulation can be generalized by introducing two linear transforms $\mathbf{F} \in \mathbb{R}^{n \times \hat{n}}$ and $\mathbf{F}^\dagger \in \mathbb{R}^{\hat{n} \times n}$:

$$\hat{\mathbf{B}} = \mathbf{F} \left((\mathbf{F}^\dagger \mathbf{X}) (\mathbf{F}^\dagger \mathbf{X})^T \circ \hat{\mathbf{L}} \right) \mathbf{F}^T \quad (5)$$

$$\delta \mathbf{x} = \mathbf{F} \sum_{j=1}^N \left(\mathbf{F}^\dagger \mathbf{X}_{:j} \right) \circ \left(\hat{\mathbf{U}} \hat{\mathbf{v}}_j \right) \quad (6)$$

where:

- $\hat{\mathbf{L}} \in \mathbb{R}^{\hat{n} \times \hat{n}}$ is the localization matrix in the transformed space.
- $\hat{\mathbf{U}} \in \mathbb{R}^{n \times \hat{m}}$ is a square-root of the localization $\hat{\mathbf{L}}$ such that $\hat{\mathbf{U}}\hat{\mathbf{U}}^T = \hat{\mathbf{L}}$.
- $\hat{\mathbf{v}}_j \in \mathbb{R}^{\hat{m}/N}$ is the j^{th} subvector of $\hat{\mathbf{v}} \in \mathbb{R}^{\hat{m}}$.

Let's analyse the asymptotic values if the ensemble size grows to infinity:

- The localization matrices $\mathbf{L}$ and $\hat{\mathbf{L}}$ should converge to $\mathbf{1}_n$ and $\mathbf{1}_{\hat{n}}$, respectively, where $\mathbf{1}_n \in \mathbb{R}^{n \times n}$ is a matrix filled with 1.
- As a consequence:

$$\lim_{N \rightarrow \infty} \mathbf{B} = \mathbf{X}\mathbf{X}^T \quad (7)$$

$$\lim_{N \rightarrow \infty} \hat{\mathbf{B}} = (\mathbf{F}\mathbf{F}^\dagger \mathbf{X}) (\mathbf{F}\mathbf{F}^\dagger \mathbf{X})^T \quad (8)$$

- To ensure consistency, linear transforms $\mathbf{F}$ and $\mathbf{F}^\dagger$ should not have an impact on the asymptotic background error covariance:

$$\lim_{N \rightarrow \infty} \mathbf{B} = \lim_{N \rightarrow \infty} \hat{\mathbf{B}} \quad (9)$$

so

$$\mathbf{F}\mathbf{F}^\dagger \mathbf{X} = \mathbf{X} \quad (10)$$

Thus, $\mathbf{F}^\dagger$ should be a right-inverse of $\mathbf{F}$ in the ensemble perturbations space.

The content of $\mathbf{F}$ and $\mathbf{F}^\dagger$ can be very diverse:

- A change of variables: for instance towards a better-suited humidity variable.

- A inverse balance operator as in ?, where the localization is applied on unbalanced variables and the balance operator is finally applied to get the increment.
- A change of resolution as in ?, where the ensemble resolution is lower than the final increment one. In this case, $\mathbf{F}$ is an interpolation from low to high resolution, and the low-resolution perturbations $\mathbf{X}_{\text{LR}}$ can be thought as the application of a coarsening operator $\mathbf{F}^\dagger$ on virtual high-resolution perturbations $\mathbf{X}_{\text{HR}}$ representing the same field:

$$\mathbf{F}\mathbf{X}_{\text{LR}} = \mathbf{F}\mathbf{F}^\dagger\mathbf{X}_{\text{HR}} = \mathbf{X}_{\text{HR}} \quad (11)$$

2 Filter banks

2.1 The band-pass framework

We define a series of P band-pass filters $\{\mathbf{H}_1, \dots, \mathbf{H}_P\}$, where $\mathbf{H}_p \in \mathbb{R}^{n \times n}$. The linear transform $\mathbf{F}^\dagger$ is defined as:

$$\mathbf{F}^\dagger = \begin{pmatrix} \mathbf{H}_1 \\ \mathbf{H}_2 \\ \vdots \\ \mathbf{H}_P \end{pmatrix} \quad (12)$$

and its left-inverse as:

$$\mathbf{F} = \begin{pmatrix} \mathbf{I}_n & \mathbf{I}_n & \dots & \mathbf{I}_n \end{pmatrix} \quad (13)$$

Thus, $\mathbf{F}\mathbf{F}^\dagger$ is the sum of the band-pass filters:

$$\mathbf{F}\mathbf{F}^\dagger = \sum_{p=1}^P \mathbf{H}_p \quad (14)$$

and the size of the decomposition space (i.e. the number of rows of $\mathbf{F}^\dagger$) is $\hat{n} = Pn$. For $1 \leq p \leq P-1$, the band-pass filters $\mathbf{H}_p$ are defined without constraint. However, to ensure that $\mathbf{F}\mathbf{F}^\dagger = \mathbf{I}_n$, the last band-pass filter $\mathbf{H}_P$ is defined as the complement of the sum of all the previous band-pass filters:

$$\mathbf{H}_P = \mathbf{I}_n - \sum_{p=1}^{P-1} \mathbf{H}_p \quad (15)$$

We also define the filtered components $\hat{\mathbf{x}}_p \in \mathbb{R}^n$ as:

$$\hat{\mathbf{x}}_p = \mathbf{H}_p \mathbf{x} \quad (16)$$

where $\mathbf{x} \in \mathbb{R}^n$ is the input vector.

2.2 Recursive construction

In this section, we build the band-pass filters $\mathbf{H}_p$ recursively from a series of $P - 1$ filters $\{\mathbf{G}_1, \dots, \mathbf{G}_{P-1}\}$, where $\mathbf{G}_p \in \mathbb{R}^{n \times n}$:

- The initial residual is given by the input state: $\bar{\mathbf{x}}_0 = \mathbf{x}$
- At each iteration $1 \leq p \leq P - 1$:
 - The filter $\mathbf{G}_p$ is applied to the previous residual $\bar{\mathbf{x}}_{p-1} \in \mathbb{R}^n$ to produce the new filtered component $\hat{\mathbf{x}}_p \in \mathbb{R}^n$:

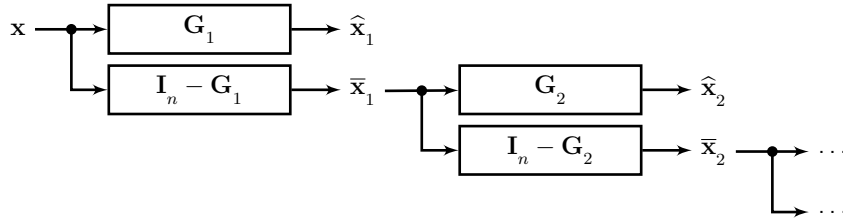
$$\hat{\mathbf{x}}_p = \mathbf{G}_p \bar{\mathbf{x}}_{p-1} \quad (17)$$

- The new residual is computed from $\hat{\mathbf{x}}_p$:

$$\bar{\mathbf{x}}_p = \bar{\mathbf{x}}_{p-1} - \hat{\mathbf{x}}_p \quad (18)$$

$$= (\mathbf{I}_n - \mathbf{G}_p) \bar{\mathbf{x}}_{p-1} \quad (19)$$

This construction can be illustrated as follows:



Integrating recursively, the residual at iteration p becomes:

$$\begin{aligned} \bar{\mathbf{x}}_p &= (\mathbf{I}_n - \mathbf{G}_p) (\mathbf{I}_n - \mathbf{G}_{p-1}) \dots (\mathbf{I}_n - \mathbf{G}_1) \mathbf{x} \\ &= \prod_{q=1}^p (\mathbf{I}_n - \mathbf{G}_q) \mathbf{x} \end{aligned} \quad (20)$$

and from (17), we obtain the filtered component at iteration p :

$$\hat{\mathbf{x}}_p = \mathbf{G}_p \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) \mathbf{x} \quad (21)$$

Combining equations (16) with (21), we get an equivalence between $\mathbf{H}_p$ and $\mathbf{G}_p$ for $1 \leq p \leq P - 1$:

$$\mathbf{H}_p = \mathbf{G}_p \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) \quad (22)$$

The last filtered component $\hat{\mathbf{x}}_P$ is actually given by the last residual $\bar{\mathbf{x}}_{P-1}$

$$\hat{\mathbf{x}}_P = \mathbf{H}_P \mathbf{x} = \bar{\mathbf{x}}_{P-1} \quad (23)$$

So equation (20) gives another expression of equation (15):

$$\mathbf{H}_P = \prod_{q=1}^{P-1} (\mathbf{I}_n - \mathbf{G}_q) \quad (24)$$

2.3 Inverse formula

To inverse equation (22), a preliminary step is necessary. We start by showing recursively that for $p > 1$:

$$\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q = \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) \quad (25)$$

The initial case $p = 2$ is obvious: since $\mathbf{H}_1 = \mathbf{G}_1$ (equation (22)), then $\mathbf{I}_n - \mathbf{H}_1 = \mathbf{I}_n - \mathbf{G}_1$. We now assume that equation (25) holds for a given $p > 1$. As a consequence:

$$\begin{aligned} \mathbf{I}_n - \sum_{q=1}^p \mathbf{H}_q &= \mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q - \mathbf{H}_p \\ &= \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) - \mathbf{G}_p \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) \\ &= (\mathbf{I}_n - \mathbf{G}_p) \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) \\ &= \prod_{q=1}^p (\mathbf{I}_n - \mathbf{G}_q) \end{aligned} \quad (26)$$

so equation (25) also holds for $p + 1$. We can conclude that (25) holds for any $p > 2$. Inserting it into equation (22), we get:

$$\mathbf{H}_p = \mathbf{G}_p \left(\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q \right) \quad (27)$$

If $\left(\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q \right)$ has a right inverse, then the filter $\mathbf{G}_p$ can be deduced from the band-pass filters:

$$\mathbf{G}_p = \mathbf{H}_p \left(\mathbf{I}_n - \sum_{q=1}^{p-1} \mathbf{H}_q \right)^{-1} \quad (28)$$

2.4 Low-pass and high-pass filters

For a sequence of filters, we assume that there is an invertible transform $\mathbf{S} \in \mathbb{R}^{n \times n}$ such that every component can be expressed as:

$$\mathbf{G}_p = \mathbf{S}^{-1} \mathbf{D}_p \mathbf{S} \quad (29)$$

where $\mathbf{D}_p \in \mathbb{R}^{n \times n}$ is a diagonal matrix with coefficients between 0 and 1. The spectrum $\mathbf{d}_p \in \mathbb{R}^n$ contains all these coefficients: $D_{p,ii} = d_{p,i}$.

For a sequence of low-pass filters, all the spectra start from 1 and are monotonically decreasing. Symmetrically for a sequence of high-pass filters, all the spectra are monotonically increasing and end up at 1. In both cases, for any $p > 1$ and any coefficient $d_{p,i}$ of $\mathbf{d}_p$: $d_{p,i} \geq d_{p-1,i}$, as illustrated in Figure 1.

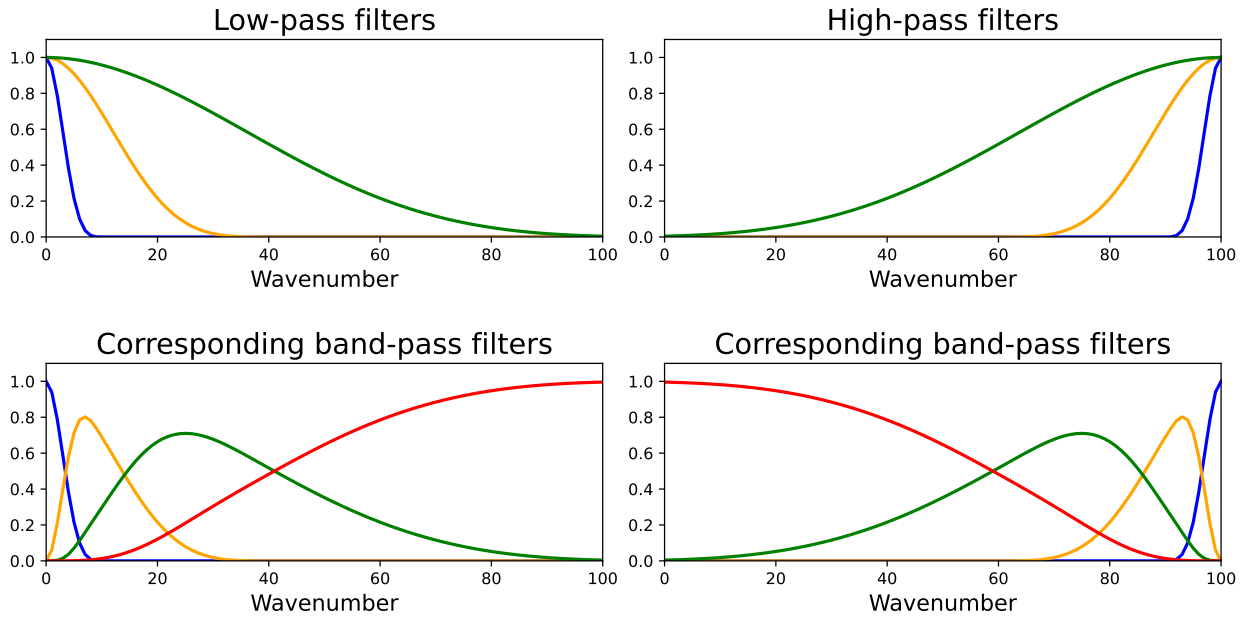


Figure 1: Spectra of low-pass filters (top left) and high-pass filters (top right), and their equivalent band-pass filters (bottom).

2.5 Embedded filters

The special case of embedded filters arises when for any $p > q$:

$$\mathbf{G}_p \mathbf{G}_q = \mathbf{G}_q \mathbf{G}_p = \mathbf{G}_q \quad (30)$$

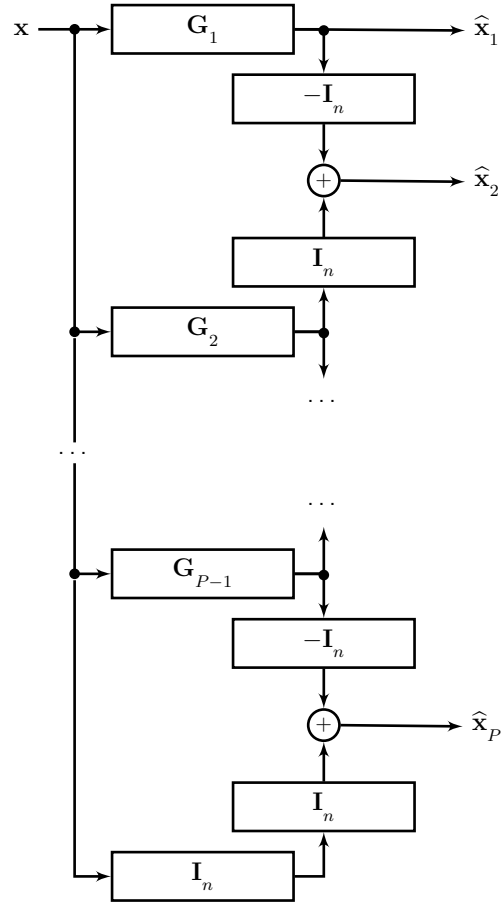
The corresponding band-pass filters can then be expressed for $1 \leq p \leq P - 1$ as:

$$\begin{aligned}
\mathbf{H}_p &= \mathbf{G}_p \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q) \\
&= (\mathbf{G}_p - \mathbf{G}_p \mathbf{G}_{p-1}) \prod_{q=1}^{p-2} (\mathbf{I}_n - \mathbf{G}_q) \\
&= (\mathbf{G}_p - \mathbf{G}_{p-1}) \prod_{q=1}^{p-2} (\mathbf{I}_n - \mathbf{G}_q) \\
&= \mathbf{G}_p - \mathbf{G}_{p-1}
\end{aligned} \tag{31}$$

because for each factor of $\prod_{q=1}^{p-2} (\mathbf{I}_n - \mathbf{G}_q)$:

$$\begin{aligned}
(\mathbf{G}_p - \mathbf{G}_{p-1}) (\mathbf{I}_n - \mathbf{G}_q) &= \mathbf{G}_p - \mathbf{G}_{p-1} - (\mathbf{G}_p \mathbf{G}_q - \mathbf{G}_{p-1} \mathbf{G}_q) \\
&= \mathbf{G}_p - \mathbf{G}_{p-1} - (\mathbf{G}_q - \mathbf{G}_q) \\
&= \mathbf{G}_p - \mathbf{G}_{p-1}
\end{aligned} \tag{32}$$

As a consequence, the construction of a sequence of embedded filters can be illustrated as follows:



If a sequence of low-pass or high-pass filters is embedded, equation (30) can be simplified into

$$d_{p,i} d_{q,i} = d_{q,i} \quad (33)$$

which is only true if:

$$\begin{cases} \text{either } d_{q,i} = 0 \\ \text{or } d_{p,i} = 1 \end{cases} \quad (34)$$

A sequence of embedded low-pass filters is illustrated on Figure 2.

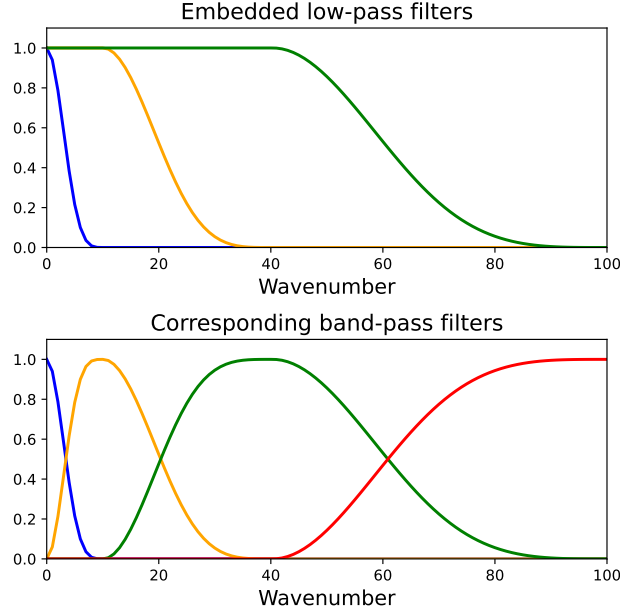


Figure 2: Spectra of embedded low-pass filters (top) and the equivalent band-pass filters (bottom).

2.6 High-pass filters built from low-pass filters

Often in practice, only low-pass filters are actually implemented, through a diffusion process or an equivalent smoother. However, the high-pass filter $\mathbf{G}_p^h \in \mathbb{R}^{n \times n}$ can be built as the complement of the low-pass filter sorted in opposite order $\mathbf{G}_{P-p}^l \in \mathbb{R}^{n \times n}$:

$$\mathbf{G}_p^h = \mathbf{I}_n - \mathbf{G}_{P-p}^l \quad (35)$$

Updating equation (22), the band-pass filter $\mathbf{H}_p^h$ corresponding to the high-pass filters is given for $1 \leq p \leq P-1$ by:

$$\begin{aligned} \mathbf{H}_p^h &= \mathbf{G}_p^h \prod_{q=1}^{p-1} (\mathbf{I}_n - \mathbf{G}_q^h) \\ &= (\mathbf{I}_n - \mathbf{G}_{P-p}^l) \prod_{q=1}^{p-1} \mathbf{G}_{P-q}^l \end{aligned} \quad (36)$$

Figure 3 shows explicit low-pass filters and their complementary high-pass filters counterparts. The equivalent band-pass filters are also displayed in both cases. We can see that the band-pass filters have roughly the same structure in both cases, but are not identical.

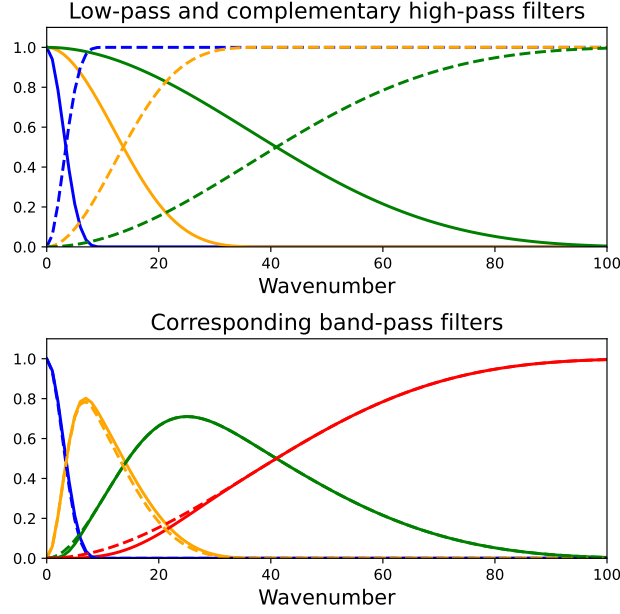


Figure 3: Spectra of explicit low-pass filters (top left) and their complementary high-pass filters counterparts (top right). The equivalent band-pass filters are displayed for both cases in the bottom panels, the dashed black line indicating the sum of all the band-pass filters.

If the sequence of low-pass filters is embedded, it is easy to see that the sequence of complementary high-pass filters is embedded too, and the corresponding band-pass filters can be derived from equation (36):

$$\begin{aligned} \mathbf{H}_p^h &= \left(\mathbf{I}_n - \mathbf{G}_{P-p}^l \right) \prod_{q=1}^{p-1} \mathbf{G}_{P-q}^l \\ &= \mathbf{G}_{P-p+1}^l - \mathbf{G}_{P-p}^l \end{aligned} \quad (37)$$

So the corresponding band-pass filters are equivalent for the low-pass filters and their high-pass complementary filters (sorted in opposite order):

$$\mathbf{H}_p^h = \mathbf{H}_{P-p+1}^l \quad (38)$$

2.7 Pseudo-embedded filters